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Autonomous and Adaptive Systems - Project

Multi-Agent Social Dilemmas

Introduction and Setup

🦌 Hunt:

hunting the stag yield high reward but cooperation is needed

🌳Harvest:

plants grow over time, mature plants give higher reward

👣Escalation:

step-by-step coordination toward the stag, otherwise penalty

Different algorithms:

- **Four DQN variants:** Replay Memory, ε-greedy, Policy and Target Network, Separate Avantage estimation.
- **A2C:** approximate policy and value function, use Advantage Function.
- **MAPPO:** Multi Agent Proximal Policy Optimization uses General Advantage Function. Centralized Training Decentralized Execution (CTDE).

Stag-Hunt Experiments and Results

Standard Configuration:

Double and Dueling DQN outperform the on-policy methods (**rare event discovery problem**).

Vanilla DQN suffers from **moving target problem**.

With **higher LRs** MAPPO has better performance (in terms of stags-hunted).

Reward Analysis:

More **permissive rewards** shapes give to the algorithms better performances (stags-hunted).

⚠️Anomaly with 10, -1

Severe reward shaping makes agents stack in **sub-optimal foraging behaviour**.

Grid sizes:

Increasing the grid size **decrease** the agents **performances**.

Dueling DQN is the most robust.

Rare event discovery problem amplified, MAPPO stack in sub-optimal risk-free.

Harvest and Escalation

Harvest:

MAPPO and A2C dominate harvest game thanks to the higher frequency of “good” event.

No algorithm learn the **delayed harvesting** behaviour in order to increase the number of total reward obtained (**greedy behaviour**), despite reward shaping (e.g., 5, -5).

Escalation:

MAPPO outperforms all other algorithm probably thanks to its **Centralized Training Decentralized Execution** paradigm.

A2C totally fails despite reward shaping (e.g., 0.1 punishment factor).

Dueling DQN is the most stable among DQNs variation.